

## SAFETY DATA SHEET

Revision Date 24-December-2021

Revision Number 5

### 1. Identification

**Product Name** (R)-(-)-sec-Butanol

**Cat No. :** AC217230000; AC217230010; AC217230050; AC217230250

**CAS-No** 14898-79-4

**Synonyms** (R)-(-)-sec-Butyl alcohol; (R)-(-)-2-Butanol

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                   |            |
|-------------------------------------------------------------------|------------|
| <b>Flammable liquids</b>                                          | Category 3 |
| <b>Serious Eye Damage/Eye Irritation</b>                          | Category 2 |
| <b>Specific target organ toxicity (single exposure)</b>           | Category 3 |
| Target Organs - Respiratory system, Central nervous system (CNS). |            |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Flammable liquid and vapor  
Causes serious eye irritation  
May cause respiratory irritation

May cause drowsiness and dizziness



### Precautionary Statements

#### Prevention

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Keep container tightly closed

Ground/bond container and receiving equipment

Use explosion-proof electrical/ventilating/lighting/equipment

Use only non-sparking tools

Take precautionary measures against static discharges

Avoid breathing dust/fume/gas/mist/vapors/spray

Wash face, hands and any exposed skin thoroughly after handling

Use only outdoors or in a well-ventilated area

Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Call a POISON CENTER/ doctor if you feel unwell

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### Storage

Store in a well-ventilated place. Keep container tightly closed

Store locked up

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component      | CAS-No     | Weight % |
|----------------|------------|----------|
| (R)-Butan-2-ol | 14898-79-4 | 99       |

## 4. First-aid measures

|                                        |                                                                                                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                      |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Get medical attention.                                                         |
| <b>Inhalation</b>                      | Remove from exposure, lie down. Remove to fresh air. If breathing is difficult, give oxygen. If not breathing, give artificial respiration. Immediate medical attention is required. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. Get medical attention. If possible drink milk afterwards.                      |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. . Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting                                                                       |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Carbon dioxide (CO <sub>2</sub> ). Foam. Water. Dry chemical. Water mist may be used to cool closed containers. water fog. Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                     |
| <b>Flash Point</b>                      | 26 °C / 78.8 °F                                                                                                                                                              |
| <b>Method -</b>                         | No information available                                                                                                                                                     |
| <b>Autoignition Temperature</b>         | 390 °C / 734 °F                                                                                                                                                              |
| <b>Explosion Limits</b>                 |                                                                                                                                                                              |
| <b>Upper</b>                            | 9.8% @ 100°C                                                                                                                                                                 |
| <b>Lower</b>                            | 1.7% @ 100°C                                                                                                                                                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                     |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                     |

### Specific Hazards Arising from the Chemical

Flammable. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Vapors may form explosive mixtures with air.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
2

**Flammability**  
2

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b> | See Section 12 for additional Ecological Information.                                  |

|                                             |                                                                                                                                                                                                                                               |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Keep away from open flames, hot surfaces and sources of ignition. Take precautionary measures against static discharges. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Keep away from heat, sparks and flame. Flammables area. Keep container tightly closed in a dry and well-ventilated place. Incompatible Materials. Acids. Halogens. Acid anhydrides. Acid chlorides.                                                                                          |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

**Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**  
**Hand Protection**

Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers recommendations | -               | Splash protection only |
| Neoprene       |                                   |                 |                        |
| Natural rubber |                                   |                 |                        |
| PVC            |                                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**9. Physical and chemical properties**

|                                        |                                         |
|----------------------------------------|-----------------------------------------|
| Physical State                         | Liquid                                  |
| Appearance                             | Colorless                               |
| Odor                                   | sweet                                   |
| Odor Threshold                         | No information available                |
| pH                                     | No information available                |
| Melting Point/Range                    | -114 °C / -173.2 °F                     |
| Boiling Point/Range                    | 99 - 100 °C / 210.2 - 212 °F @ 760 mmHg |
| Flash Point                            | 26 °C / 78.8 °F                         |
| Evaporation Rate                       | No information available                |
| Flammability (solid,gas)               | Not applicable                          |
| Flammability or explosive limits       |                                         |
| Upper                                  | 9.8% @ 100°C                            |
| Lower                                  | 1.7% @ 100°C                            |
| Vapor Pressure                         | 17 hPa @ 20.0 °C                        |
| Vapor Density                          | 2.6 (Air = 1.0)                         |
| Specific Gravity                       | 0.800                                   |
| Solubility                             | No information available                |
| Partition coefficient; n-octanol/water | No data available                       |
| Autoignition Temperature               | 390 °C / 734 °F                         |

|                           |                          |
|---------------------------|--------------------------|
| Decomposition Temperature | No information available |
| Viscosity                 | 4.21 mPa s at 20 °C      |
| Molecular Formula         | C4 H10 O                 |
| Molecular Weight          | 74.12                    |

## 10. Stability and reactivity

|                                  |                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                               |
| Stability                        | Stable under normal conditions.                                                          |
| Conditions to Avoid              | Incompatible products. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Acids, Halogens, Acid anhydrides, Acid chlorides                                         |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> )                                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                 |
| Hazardous Reactions              | None under normal processing.                                                            |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

**Component Information**  
**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component      | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------|------------|------------|------------|------------|------------|------------|
| (R)-Butan-2-ol | 14898-79-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Central nervous system (CNS)

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

**Ecotoxicity**

Do not empty into drains.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.**Bioaccumulation/ Accumulation** No information available.**Mobility** Will likely be mobile in the environment due to its water solubility.**13. Disposal considerations****Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.**14. Transport information****DOT**

|                      |          |
|----------------------|----------|
| UN-No                | UN1120   |
| Proper Shipping Name | BUTANOLS |
| Hazard Class         | 3        |
| Packing Group        | III      |

**TDG**

|                      |          |
|----------------------|----------|
| UN-No                | UN1120   |
| Proper Shipping Name | BUTANOLS |
| Hazard Class         | 3        |
| Packing Group        | III      |

**IATA**

|                      |          |
|----------------------|----------|
| UN-No                | UN1120   |
| Proper Shipping Name | BUTANOLS |
| Hazard Class         | 3        |
| Packing Group        | III      |

**IMDG/IMO**

|                      |          |
|----------------------|----------|
| UN-No                | UN1120   |
| Proper Shipping Name | BUTANOLS |
| Hazard Class         | 3        |
| Packing Group        | III      |

**15. Regulatory information****International Inventories**

| Component      | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|----------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| (R)-Butan-2-ol | 14898-79-4 | -   | -    | -    | -                                             | 238-967-8 | -      | -   |

| Component      | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------|------------|-------|----------|------|------|------|------|-------|-------|
| (R)-Butan-2-ol | 14898-79-4 | X     | KE-03871 | X    | X    | X    | -    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances**IECSC** - Chinese Inventory of Existing Chemical Substances**KECL** - Korean Existing and Evaluated Chemical Substances**ENCS** - Japanese Existing and New Chemical Substances**AICS** - Australian Inventory of Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

##### Authorisation/Restrictions according to EU REACH

| Component      | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| (R)-Butan-2-ol | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

##### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component      | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| (R)-Butan-2-ol | 14898-79-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component      | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| (R)-Butan-2-ol | 14898-79-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Revision Summary** This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**